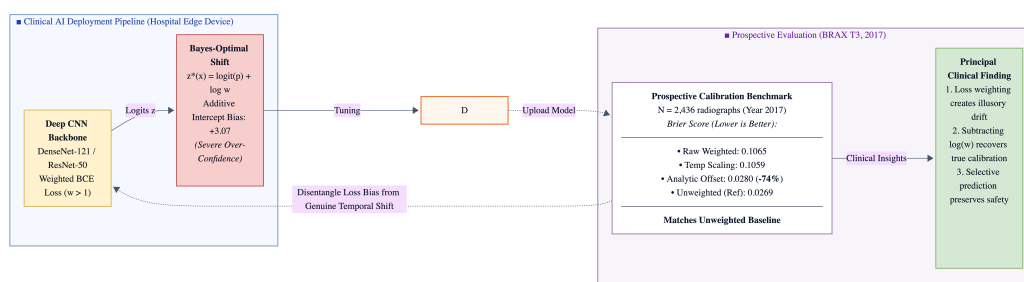


Graphical Abstract

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Temporal Cohort Shift

Vishesh Panghal



Highlights

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- Positive loss weighting induces an additive population logit offset of $\log w$.
- Finite neural network logits empirically cluster tightly around theoretical $\log w$.
- Temperature scaling cannot correct this intercept-type calibration bias.
- Analytic offset ($z - \log w$) cuts 74% of Brier error without retraining.
- Selective triage exposes trade-offs between calibrated risk and screening referral.

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Temporal Cohort Shift

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Abstract

Clinical deep learning models frequently exhibit calibration decay under longitudinal deployment. Under extreme class imbalance, training with positive-weighted binary cross-entropy (BCE) is standard practice, but its downstream effect on probability calibration has been widely conflated with intrinsic feature drift. We resolve this conflation through a controlled $2 \times 2 \times 3$ factorial experiment across 12 models (DenseNet-121 and ResNet-50 trained under unweighted and positive-weighted BCE across 3 seeds) on 40,523 chest radiographs from BRAX. Models are trained on an anchor partition and evaluated across chronologically subsequent deidentified-date strata (T_1, T_2, T_3) with strict patient isolation. We show that positive loss weighting mathematically induces an additive population logit shift of $\log w$, which temperature scaling structurally cannot correct. In finite networks, empirical logit differences cluster tightly around theoretical $\log w$ (mean $\Delta z = 3.29 \pm 0.81$ vs. theoretical 3.07 for Pleural Effusion). On stratum T_3 , zero-parameter analytic offset ($z - \log w$) eliminates approximately 74% of weighted BCE Brier error ($\Delta \text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]), driving calibration intercept from $\alpha = -2.96$ to -0.45 without retraining or validation data. On an external MIMIC-CXR-JPG cohort (3,403 radiographs, 289 patients), discrimination proved pathology-dependent (Pleural Effusion AUROC 0.688; Cardiomegaly 0.523), with severe baseline calibration shift ($\text{BSS} < 0$). Analytic offset yielded significant Brier reductions for Cardiomegaly and Pneumonia, but

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worsened Edema, delineating boundaries of zero-parameter recalibration under cross-institutional shift. Finally, selective prediction simulations expose operational trade-offs between risk estimation and high-sensitivity screening triage.

Keywords: Chest radiography, Probability calibration, Class imbalance, Distribution shift, Selective prediction, Deep learning, Clinical decision support

1. Introduction

Deep learning models for automated chest radiograph interpretation have achieved high discriminatory performance across common thoracic abnormalities such as pleural effusion, cardiomegaly, and pneumonia [1, 2]. However, the transition from stationary retrospective validation to longitudinal clinical deployment introduces substantial distribution shifts. In operational healthcare environments, patient populations evolve, imaging protocols shift, scanner hardware is replaced, and demographic or seasonal patterns alter condition prevalence over time [3, 4, 5].

While discriminatory ranking (measured via AUROC) often exhibits gradual longitudinal attenuation under temporal shift, decision-critical applications rely fundamentally on *calibrated posterior probabilities*: $P(Y = 1 \mid X = x)$ [6, 7]. When predicted probabilities deviate from true empirical event rates, downstream operational decision cutoffs shift unpredictably, expected utility formulations degrade, and automated decision-support pipelines generate distorted risk rankings.

Recent literature has reported rapid “temporal calibration decay,” asserting that frozen deep neural networks suffer progressive miscalibration over deployment periods [8, 9]. Simultaneously, because biomedical datasets are characterized by extreme class imbalance (pathology prevalence often below 5%), researchers routinely train vision models using positive-weighted binary cross-entropy (BCE) to penalize missed positive cases [1, 10].

In this work, we present a computational investigation showing that these two phenomena are deeply entangled, and that the prevailing attribution of temporal calibration degradation primarily to intrinsic feature drift may be largely explained by uncorrected post-training calibration offsets under label distribution shifts [11, 12]. Under population-risk minimization, when models are trained with positive loss weighting $w > 1$, the Bayes-optimal

logit is shifted by an additive constant $+\log w$ [13]. Standard post-hoc calibration methods—most notably temperature scaling [14]—only rescale the logit variance (z/T) and cannot shift the logit intercept. Consequently, weighted models continuously over-predict positive probabilities, producing severe Brier score and Expected Calibration Error (ECE) degradation that compounds over time as disease prevalence fluctuates.

1.1. Related Work

Probability Calibration in Deep Learning: Modern deep convolutional neural networks frequently exhibit over-confidence due to unconstrained cross-entropy minimization with over-parameterized architectures [14]. Post-hoc recalibration techniques such as temperature scaling, histogram binning, and isotonic regression have been widely evaluated [14, 15]. In medical decision models, calibration accuracy is paramount for valid risk stratification [6, 7]. However, standard methods were primarily derived for multi-class classification under balanced distributions, leaving binary imbalanced settings less formally explored.

Class Imbalance and Loss Function Design: Extreme class imbalance is pervasive in thoracic imaging. Common mitigation strategies include positive loss weighting [10, 1], focal loss [16], and re-sampling. While positive weighting successfully shifts operational sensitivity, King and Zeng [13] proved in classical logistic regression that sample proportion adjustments shift the model intercept by $\log w$. Modern deep learning pipelines routinely apply loss weighting without correcting this induced logit offset prior to post-hoc calibration.

Distribution Shift and Label Shift Adaptation: Temporal dataset shift encompasses changes in feature distributions $P(X)$, conditional distributions $P(Y | X)$, and prior label probabilities $P(Y)$ [3]. Davis et al. [12] demonstrated that calibration drift often precedes discriminatory loss in clinical models. Alexandari et al. [11] established that maximum likelihood with bias-corrected calibration provides robust adaptation under label shift. Decoupling algorithmic training offsets from genuine temporal shift is essential for diagnostic integrity.

Selective Prediction and Deferral Mechanisms: Selective classification allows models to abstain from predicting on ambiguous inputs, routing them to human experts [17, 18]. Fisch et al. [19] highlighted that selective classification performance is fundamentally bounded by probability calibration. Investigating how post-hoc calibration interacts with uncertainty-based

66 triage under temporal shift addresses an open operational question.

67 *1.2. Contributions*

68 To disentangle algorithmic loss bias from longitudinal distribution shift,
69 this paper provides:

- 70 1. **A Controlled Factorial Matrix:** We execute a $2 \times 2 \times 3$ factorial
71 design across 12 models (DenseNet-121 and ResNet-50 architectures $\times$
72 standard BCE and positive-weighted BCE $\times$ 3 random seeds) trained
73 on 25,123 chest radiographs and evaluated on 8,302 chronologically
74 held-out radiographs across three sequential strata (T_1, T_2, T_3) from the
75 BRAX dataset [2].
- 76 2. **Theoretical Derivation and Empirical Verification of Logit**
77 **Shift:** Under population risk minimization, we derive that positive-
78 weighted BCE introduces a constant logit offset $\log w$. In finite neural
79 networks, we verify empirically that observed logit differences ($z_w - z_u$)
80 tightly mirror theoretical $\log w$ across multiple thoracic findings. We
81 show that temperature scaling cannot adjust the logit intercept, leaving
82 the calibration intercept severely biased ($\alpha \approx -2.96$), whereas an exact
83 zero-parameter analytic offset ($z - \log w$) or non-negative Platt scaling
84 restores calibration parity with unweighted models ($\alpha \rightarrow -0.45$, Brier
85 error reduced by 74%).
- 86 3. **A Leak-Free Prospective Calibration Protocol:** We implement
87 a strict 5-fold patient-clustered cross-fitting protocol on T_1 , strictly
88 preventing in-sample calibration leakage, and report prospective gen-
89 eralization on T_2 and T_3 backed by 2,000-replicate patient-clustered
90 bootstrap confidence intervals directly on individual seeds and the 3-seed
91 ensemble.
- 92 4. **Human-in-the-Loop Selective Prediction Simulation:** We for-
93 mulate a selective triage simulation with explicit, transparent tracking
94 of reader referral workloads, retained sensitivity ($TP/N_{\text{pos, ret}}$), and
95 full-cohort system sensitivity ($1 - FN/N_{\text{pos, total}}$) under matched de-
96 velopment thresholds, demonstrating that calibrated models optimize
97 probability estimation while raw weighted models support recall-oriented
98 screening triage.

2. Materials and Methods

2.1. Dataset and Longitudinal Cohort Stratification

We utilize the Brazilian Chest X-ray (BRAX) v1.1.0 benchmark [2], containing 40,523 chest radiographs acquired across multiple inpatient and outpatient emergency facilities at Hospital Israelita Albert Einstein in São Paulo, Brazil. Radiographs are labeled with 14 thoracic findings extracted from Portuguese radiological reports using a validated natural language processing pipeline.

In compliance with HIPAA de-identification standards, direct patient identifiers and original timestamps are protected. We extract chronological study dates from the released 8-digit calendar integers (YYYYMMDD), spanning 2008 through 2017. As is standard under HIPAA Safe Harbor de-identification, date anonymization may include patient-specific random date shifting that preserves relative within-patient temporal intervals while shifting absolute calendar offsets across patients. Consequently, our prospective partitions T_1 (2015-H2), T_2 (2016), and T_3 (2017) are interpreted as chronologically ordered cohort strata under longitudinal distribution shift rather than synchronized real-world calendar years. To maintain strict prospective integrity and preclude longitudinal data leakage:

1. All studies are partitioned chronologically around a strict model freeze boundary: $T_{\text{freeze}} = 2015-07-01$.
2. All training studies occur before 2013-09-01 ($N = 25,123$ images from 11,265 patients across 14,315 studies).
3. All validation studies occur between 2013-09-01 and 2015-06-30 ($N = 7,098$ images from 3,281 patients across 4,012 studies).
4. All prospective test studies occur between 2015-07-01 and 2017-12-17 ($N = 8,302$ images from 3,807 patients across 4,670 studies), partitioned into three consecutive deployment strata: T_1 (2015-H2, $N = 1,879$), T_2 (2016, $N = 3,987$), and T_3 (2017, $N = 2,436$).
5. Strict patient isolation is enforced: $\text{Patients}_{\text{train}} \cap \text{Patients}_{\text{val}} \cap \text{Patients}_{\text{test}} = \emptyset$. A small subset of 89 patients whose visits crossed split boundaries were excluded from validation and testing to guarantee zero within-patient contamination.
6. For longitudinal paired logit offset and multi-calibrator evaluation, all 8,302 prospective radiographs ($N = 1,879$ in T_1 , $N = 3,987$ in T_2 , and $N = 2,436$ in T_3) are evaluated across both clinical architectures

Table 1: Cohort Demographics, Radiographic Projections, and Target Condition Prevalence across Stratified Cohorts

Cohort Stratum	Images (N)	Patients (N)	Studies (N)	Sex (% M / F)	Age (Mean $\pm$ SD)	Views (% PA / AP / Lat / Unk)	Pleural Effusion N (%)	Cardiomegaly N (%)	Pneumonia N (%)	Edema N (%)
Train (Pre-deployment)	25,123	11,265	14,315	49.3% / 50.7%	46.0 $\pm$ 24.3	34.0% / 13.5% / 0.0% / 27.2%	1,116 (4.44%)	2,398 (9.55%)	456 (1.82%)	23 (0.09%)
Val (Pre-deployment)	7,098	3,281	4,012	49.8% / 50.2%	46.2 $\pm$ 24.1	34.7% / 12.2% / 0.0% / 27.7%	295 (4.16%)	681 (9.59%)	128 (1.80%)	8 (0.11%)
Test T1 (2015)	1,879	913	1,039	47.8% / 52.2%	44.5 $\pm$ 23.9	34.4% / 10.2% / 0.0% / 29.0%	99 (5.27%)	147 (7.82%)	50 (2.66%)	4 (0.21%)
Test T2 (2016)	3,987	1,832	2,249	48.5% / 51.5%	46.1 $\pm$ 23.8	34.6% / 12.0% / 0.0% / 27.6%	177 (4.44%)	422 (10.58%)	79 (1.98%)	13 (0.33%)
Test T3 (2017)	2,436	1,164	1,382	50.9% / 49.1%	46.0 $\pm$ 24.6	34.4% / 12.4% / 0.0% / 28.1%	85 (3.49%)	258 (10.59%)	50 (2.05%)	0 (0.00%)
Test All (Pooled)	8,302	3,807	4,670	49.0% / 51.0%	45.7 $\pm$ 24.1	34.5% / 11.7% / 0.0% / 28.1%	361 (4.35%)	827 (9.96%)	179 (2.16%)	17 (0.20%)

(DenseNet-121 and ResNet-50) with zero prospective exclusions (0% exclusion rate), yielding 16,604 paired model evaluations ($8,302 \times 2$) with complete longitudinal tracking.

Demographic, projection, and disease prevalence characteristics across all cohort strata are detailed in Table 1.

2.2. External Transport Validation Cohort (MIMIC-CXR-JPG)

To evaluate cross-institutional transportability under realistic clinical domain shift, frozen BRAX-trained models were externally evaluated on all eligible frontal radiographs (AP and PA views) from the official patient-partitioned MIMIC-CXR-JPG test partition [20, 21]. The cohort comprises $N = 3,403$ frontal radiographs ($N = 3,041$ diagnostic studies across $N = 289$ unique patients; AP projections: 2,405 [70.67%], PA projections: 998 [29.33%]). In strict accordance with the primary evaluation protocol, ground-truth diagnostic labels were extracted using the official CheXpert labeler under the standard uncertainty-zero (U -zero) policy. Empirical condition prevalence across images was: Pleural Effusion 32.18% ($N = 1,095$ positive radiographs), Cardiomegaly 26.33% ($N = 896$), Edema 21.33% ($N = 726$), and Pneumonia 10.05% ($N = 342$). Because multiple radiographs and studies belong to the same patient, patient-clustered non-parametric bootstrap inference was used throughout external evaluation to account for repeated within-patient observations, avoiding artificial inflation of effective sample size. All 3,403 radiographs were physically verified on disk (100.00% completeness). No retraining or hyperparameter fine-tuning was performed, isolating out-of-distribution transportability from in-domain adaptation.

2.3. Architectures and Training Objectives

We evaluate two standard clinical vision architectures initialized from ImageNet pre-trained weights:

- **DenseNet-121** [22]: Employs dense feature concatenation, standard for thoracic pathology benchmarks [1].

164 • **ResNet-50** [23]: Employs residual skip connections as an independent
 165 structural comparator.

166 Each architecture is trained under two distinct loss objectives:

167 1. **Unweighted Binary Cross-Entropy (Standard BCE):**

$$\mathcal{L}_{\text{std}}(y, \hat{p}) = -[y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (1)$$

168 where $\hat{p} = \sigma(z) = 1/(1 + e^{-z})$ is the sigmoid probability from raw logit
 169 z .

170 2. **Positive-Weighted Binary Cross-Entropy (Weighted BCE):**

$$\mathcal{L}_{\text{pos}}(y, \hat{p}) = -[w \cdot y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (2)$$

171 where $w = (N_{\text{total}} - N_{\text{pos}})/N_{\text{pos}}$ represents the inverse positive prevalence
 172 in the training set (e.g., $w = 21.51$ for Pleural Effusion, $w = 9.48$ for
 173 Cardiomegaly, $w = 54.09$ for Pneumonia).

174 Models are trained with the AdamW optimizer (initial learning rate 10^{-4} ,
 175 weight decay 10^{-2}) and a Cosine Annealing learning rate schedule over 15
 176 epochs, with batch size 64 and Automatic Mixed Precision (AMP fp16) on
 177 512×512 pixel input resolution. Checkpoints are selected based on the epoch
 178 achieving the highest validation Pleural Effusion AUROC across the 15 epochs
 179 (occurring between epochs 4 and 15 across runs), without early stopping
 180 loop termination. Each configuration is executed across three random seeds
 181 ($S \in \{42, 123, 2026\}$), producing 12 fully independent models. For unified
 182 system-level benchmarking, a 3-seed probability ensemble is constructed for
 183 each architecture and loss objective by averaging predicted probabilities across
 184 independent training runs:

$$\bar{p}_{\text{raw}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x)), \quad \bar{p}_{\text{corrected}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x) - \log w) \quad (3)$$

185 All post-hoc calibration, paired analytic adjustments, and selective prediction
 186 policies are evaluated directly on both individual model seeds and the unified
 187 probability ensemble, ensuring strict mathematical consistency across all
 188 reported metrics and tables.

189 *2.4. Mathematical Mechanics of Loss Weighting and Calibration*

190 Under positive-weighted binary cross-entropy, the empirical risk minimizer
191 seeks to minimize the expected loss:

$$\begin{aligned}\mathbb{E}_{Y|X}[\mathcal{L}_{\text{pos}}(Y, \hat{p})] &= -w \cdot P(Y = 1 | X) \log \hat{p} \\ &\quad - P(Y = 0 | X) \log(1 - \hat{p})\end{aligned}\tag{4}$$

192 Setting the derivative with respect to logit z to zero:

$$\begin{aligned}\frac{\partial}{\partial z} \mathbb{E}[\mathcal{L}_{\text{pos}}] &= \hat{p} [wP(Y = 1|X) + P(Y = 0|X)] \\ &\quad - wP(Y = 1|X) = 0\end{aligned}\tag{5}$$

193 Solving for the optimal predicted probability $\hat{p}^*$:

$$\hat{p}^* = \frac{w \cdot P(Y = 1 | X)}{w \cdot P(Y = 1 | X) + P(Y = 0 | X)}\tag{6}$$

194 Converting to log-odds:

$$z^*(X) = \log \left(\frac{\hat{p}^*}{1 - \hat{p}^*} \right) = \log \left(\frac{P(Y = 1 | X)}{P(Y = 0 | X)} \right) + \log w\tag{7}$$

195 Equation (7) derives that positive loss weighting shifts the model’s population
196 Bayes-optimal logit by $+\log w$. When $w \gg 1$, the network is mathematically
197 incentivized to output large positive logits, systematically inflating predicted
198 probabilities. We emphasize that Equation (7) describes the population-risk
199 minimizer under infinite capacity; in finite neural networks trained with
200 stochastic gradient descent, empirical logits fluctuate around this theoretical
201 attractor.

202 *2.4.1. Limitation of Temperature Scaling*

203 Temperature scaling [14] adjusts probabilities via a single positive scalar
204 T :

$$p_{\text{temp}} = \sigma \left(\frac{z}{T} \right)\tag{8}$$

205 To formally evaluate calibration structure, we examine the standard logistic
206 calibration regression (Cox calibration) [6, 7]:

$$\text{logit}(P(Y = 1 | X)) = \alpha + \beta \cdot \text{logit}(\hat{p})\tag{9}$$

where $\alpha = 0$ represents perfect calibration-in-the-large (calibration intercept), and $\beta = 1$ denotes ideal calibration slope. Because positive loss weighting induces an additive shift $z \approx z_{\text{true}} + \log w$, the uncorrected logit suffers a massive intercept discrepancy ($\alpha \approx -\log w \ll 0$). Temperature scaling multiplies the logit by $1/T$, adjusting the slope $\beta \rightarrow \beta \cdot T$, but is mathematically incapable of shifting the additive constant α . In low-prevalence clinical domains where negative cases vastly dominate, the negative log-likelihood optimizer is constrained by the conflicting pressures of slope and intercept, typically selecting $T^* \approx 1.0$ or providing limited, architecture-dependent variance shrinkage without eliminating the severe underlying intercept displacement.

2.4.2. Post-Hoc Recalibration Suite

To systematically rectify this mechanism, we evaluate four post-hoc recalibration functions:

1. **Temperature Scaling (T^*):** $p = \sigma(z/T^*)$, optimized via bounded Brent minimization on $[0.01, 50.0]$ with an explicit monotonicity constraint (if $\text{NLL}(T^*) > \text{NLL}(1.0) \implies T^* = 1.0$).
2. **Analytic Offset ($z - \log w$):** Directly subtracts the theoretical training logit shift:

$$p_{\text{analytic}} = \sigma(z - \log w) \quad (10)$$

Requires zero validation data and zero numerical optimization, resetting $\alpha \approx 0$ by construction while preserving model discrimination and slope.

3. **Fitted Offset ($z + a^*$):** Optimizes a single scalar intercept $a^* \in [-10.0, 10.0]$ via univariate Brent minimization on validation log-likelihood, holding the slope fixed at 1.0.
4. **Platt Scaling ($a + bz$):** Logistic regression on logits $p = \sigma(a + bz)$ with a non-negative constraint on slope ($b \geq 0$) to prevent spurious rank reversal.

2.4.3. Algorithmic and Computational Complexity

The evaluated post-hoc calibrators exhibit distinct computational profiles:

- **Temperature Scaling:** Incurs $O(N_{\text{val}})$ scalar optimization over validation logits and $O(1)$ inference per prediction.
- **Analytic Offset:** Incurs strictly $O(1)$ closed-form scalar subtraction ($z - \log w$), requiring zero validation storage, zero parameter fitting, and $O(1)$ inference.

- 240 • **Fitted Offset:** Incurs $O(N_{\text{val}})$ univariate Brent optimization on valida-
241 tion log-likelihood and $O(1)$ inference.
- 242 • **Platt Scaling:** Incurs $O(k \cdot N_{\text{val}})$ iteratively reweighted least squares
243 fitting (where k is the number of optimization steps) and $O(1)$ inference.

244 The analytic offset thus provides an operational advantage for zero-compute
245 edge deployment, eliminating the requirement to retain or access historical
246 validation datasets.

247 2.5. Cross-Fitting and Resampling Evaluation Protocol

248 To ensure zero optimistic in-sample bias:

- 249 • On deployment stratum T_1 (2015-H2), calibrator parameters are esti-
250 mated using **5-fold patient-clustered cross-fitting**. Calibrators are
251 fit on 4 folds and evaluated out-of-fold on the 5th fold, with all studies
252 from any given patient assigned to a single fold.
- 253 • For prospective evaluation on T_2 (2016) and T_3 (2017), the calibrators
254 fit on T_1 are frozen and applied strictly out-of-sample.
- 255 • **Statistical Inference:** All 95% confidence intervals (CIs) are com-
256 puted using $B = 2,000$ patient-clustered bootstrap resamples with
257 replacement. For paired calibrator comparisons (e.g., Analytic Offset
258 vs. Raw), differences are computed on identical bootstrap resamples.

259 2.6. Selective Prediction and Deferral Policy

260 In operational deployment, models need not provide an autonomous pre-
261 diction for every image. An uncertain model can selectively defer ambiguous
262 examinations to expert human readers [17, 18].

263 To ensure fair comparisons across calibrators operating on different proba-
264 bility scales, we determine an operating decision cutoff t^* for each method
265 strictly on development stratum T_1 using Youden’s J index ($J = \text{Sensitivity} +$
266 $\text{Specificity} - 1$), without using prospective evaluation labels. For unweighted
267 models, $t^* \approx 0.014$; for analytic offset calibrated models, $t^* \approx 0.022$; and for
268 raw weighted models, $t^* \approx 0.33$. We define a threshold-relative uncertainty
269 metric based on piecewise distance from the operational cutoff t^* :

$$u(x) = 1 - \frac{|\hat{p}(x) - t^*|}{\mathbb{I}(\hat{p}(x) \geq t^*)(1 - t^*) + \mathbb{I}(\hat{p}(x) < t^*)t^*} \in [0, 1] \quad (11)$$

where $u(x) \rightarrow 1$ indicates maximal ambiguity ($\hat{p}(x) \approx t^*$) and $u(x) \rightarrow 0$ indicates decisive confidence. Crucially, because t^* scales with baseline disease prevalence (e.g., $t^* \approx 0.022$ for analytic calibrated models), predictions below t^* are normalized by t^* . Consequently, low-probability false negatives ($\hat{p} \ll t^*$) yield low uncertainty and are retained in the automated pool rather than referred. At coverage level $C \in [0.5, 1.0]$, the automated system retains the fraction C of cases with lowest uncertainty ($N_{\text{auto}} = \lfloor C \cdot N \rfloor$) and refers the remaining fraction $1 - C$ ($N_{\text{ref}} = N - N_{\text{auto}}$) to radiologist review.

To prevent metric conflation and provide complete arithmetic transparency, we evaluate triage outcomes using distinct, explicit definitions:

- **Retained Positives ($N_{\text{pos, ret}}$):** The number of true positive cases remaining in the automated cohort after ambiguous examinations have been referred to human readers.
- **Automated True Positives (TP_{auto}):** True pathology cases correctly flagged as positive by the autonomous system ($TP_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 1)$).
- **Automated False Negatives (FN_{auto} , Missed Cases):** True pathology cases erroneously classified as negative by the autonomous system ($FN_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 0)$). Note that $TP_{\text{auto}} + FN_{\text{auto}} = N_{\text{pos, ret}}$.
- **Retained Sensitivity:** Diagnostic sensitivity evaluated strictly among the retained automated decisions: $\text{Sens}_{\text{retained}} = \frac{TP_{\text{auto}}}{N_{\text{pos, ret}}} = \frac{TP_{\text{auto}}}{TP_{\text{auto}} + FN_{\text{auto}}}$.
- **Full-Cohort Triage Sensitivity:** System-wide detection sensitivity across all $N_{\text{pos, total}}$ cases in the evaluation stratum (e.g., $N_{\text{pos, total}} = 85$ in T_3), assuming that all examinations deferred to human readers receive expert diagnostic review and are correctly diagnosed: $\text{Sens}_{\text{triage}} = \frac{TP_{\text{auto}} + N_{\text{pos, ref}}}{N_{\text{pos, total}}} = 1 - \frac{FN_{\text{auto}}}{N_{\text{pos, total}}}$.
- **Referral Workload:** The fraction and count of examinations routed to readers ($N_{\text{ref}} = (1 - C) \cdot N$).

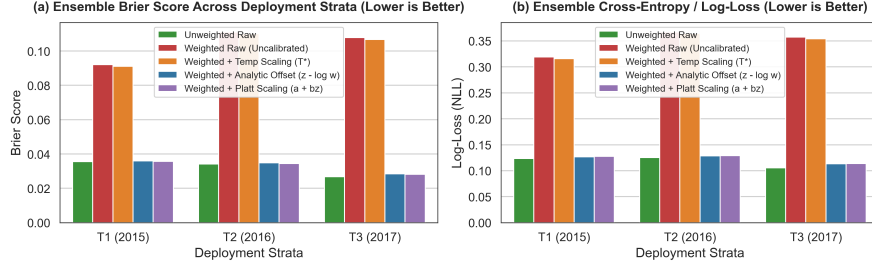


Figure 1: Prospective multi-calibrator evaluation on Pleural Effusion for the DenseNet-121 3-seed ensemble across cohorts (T_1, T_2, T_3), directly visualizing Table 3. (a) Brier score and (b) Log-loss. Temperature scaling fails to reduce probability error in weighted models, whereas Analytic Offset ($z - \log w$) and Platt Scaling eliminate 74% of Brier error, matching unweighted models.

3. Experimental Results

3.1. Factorial Matrix: Discrimination vs. Calibration Trajectories

Table 2 presents the comprehensive evaluation of the $2 \times 2 \times 3$ factorial matrix across DenseNet-121 and ResNet-50 for Pleural Effusion, Cardiomegaly, and Pneumonia across chronological strata T_1, T_2, T_3 .

3.1.1. Discrimination Attenuation

Both architectures exhibit moderate longitudinal attenuation in AUROC across chronologically subsequent cohorts. For DenseNet-121 on Pleural Effusion, AUROC decreases from 0.9335 ± 0.0052 at T_1 to 0.8731 ± 0.0120 at T_3 ($\Delta\text{AUROC} = -0.0604$). ResNet-50 exhibits an identical downward trajectory ($0.9295 \pm 0.0113 \rightarrow 0.8608 \pm 0.0190$, $\Delta\text{AUROC} = -0.0687$). This confirms that moderate discriminatory attenuation occurs across subsequent cohort strata, regardless of the training loss objective.

3.1.2. The Calibration Divergence and Factorial Interaction

In stark contrast to discrimination, probability calibration diverges completely depending on the training loss function:

- **Unweighted Models:** Brier scores are exceptionally low and actually *improve* prospectively over time for conditions with falling prevalence. For unweighted DenseNet-121 on Pleural Effusion, the Brier score improves from 0.0369 ± 0.0029 (T_1) to 0.0279 ± 0.0009 (T_3), while Equal-Mass ECE decreases from 0.0232 to 0.0090.

Table 2: Factorial Matrix Evaluation: Longitudinal Trajectories across Architectures, Loss Objectives, and Thoracic Pathologies. Values represent mean $\pm$ standard deviation across three independently trained random seeds ($S \in \{42, 123, 2026\}$).

Architecture	Objective	Target	AUROC T1	AUROC T2	AUROC T3	AUROC Δ	Brier T1	Brier T2	Brier T3	Brier Δ	ECE T1	ECE T3	ECE Δ	LogLoss T1	LogLoss T3	LogLoss Δ
DenseNet-121	Unweighted BCE	Pleural Effusion	0.9335 $\pm$ 0.0052	0.8805 $\pm$ 0.0115	0.8731 $\pm$ 0.0120	-0.0604	0.0309 $\pm$ 0.0029	0.0356 $\pm$ 0.0009	0.0279 $\pm$ 0.0009	-0.0080	0.0232 $\pm$ 0.0078	0.0090 $\pm$ 0.0022	-0.0143	0.1318 $\pm$ 0.0076	0.1114 $\pm$ 0.0027	-0.0204
DenseNet-121	Unweighted BCE	Cardiomegaly	0.9075 $\pm$ 0.0116	0.8965 $\pm$ 0.0140	0.9017 $\pm$ 0.0170	-0.0057	0.0532 $\pm$ 0.0012	0.0677 $\pm$ 0.0021	0.0677 $\pm$ 0.0028	+0.0145	0.0140 $\pm$ 0.0009	0.0210 $\pm$ 0.0074	+0.0071	0.1798 $\pm$ 0.0072	0.2231 $\pm$ 0.0102	+0.0433
DenseNet-121	Unweighted BCE	Pneumonia	0.8473 $\pm$ 0.0129	0.8818 $\pm$ 0.0106	0.8647 $\pm$ 0.0172	+0.0174	0.0235 $\pm$ 0.0008	0.0182 $\pm$ 0.0003	0.0157 $\pm$ 0.0006	-0.0048	0.0142 $\pm$ 0.0011	0.0095 $\pm$ 0.0016	-0.0046	0.0998 $\pm$ 0.0035	0.0888 $\pm$ 0.0039	-0.0107
DenseNet-121	Unweighted BCE	Edema*	0.5402 $\pm$ 0.0467	0.8640 $\pm$ 0.0686	-	-	0.0021 $\pm$ 0.0000	0.0032 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	-0.0021	0.0029 $\pm$ 0.0013	0.0029 $\pm$ 0.0029	-0.0000	0.0103 $\pm$ 0.0008	0.0029 $\pm$ 0.0029	-0.0134
DenseNet-121	Weighted BCE	Pleural Effusion	0.9601 $\pm$ 0.0537	0.8676 $\pm$ 0.0457	0.8391 $\pm$ 0.0657	-0.0670	0.1081 $\pm$ 0.0394	0.1287 $\pm$ 0.0432	0.1254 $\pm$ 0.0456	+0.0173	0.1964 $\pm$ 0.0702	0.2227 $\pm$ 0.0765	+0.0264	0.3646 $\pm$ 0.1179	0.4100 $\pm$ 0.1279	+0.0454
DenseNet-121	Weighted BCE	Cardiomegaly	0.8739 $\pm$ 0.0540	0.8589 $\pm$ 0.0600	0.8575 $\pm$ 0.0609	-0.0164	0.1262 $\pm$ 0.0247	0.1358 $\pm$ 0.0330	0.1300 $\pm$ 0.0306	+0.0088	0.2031 $\pm$ 0.0507	0.1969 $\pm$ 0.0486	-0.0062	0.3862 $\pm$ 0.0807	0.4241 $\pm$ 0.0906	+0.0379
DenseNet-121	Weighted BCE	Pneumonia	0.8181 $\pm$ 0.0206	0.8494 $\pm$ 0.0313	0.8424 $\pm$ 0.0496	+0.0243	0.1313 $\pm$ 0.0533	0.1308 $\pm$ 0.0555	0.1298 $\pm$ 0.0513	-0.0015	0.2393 $\pm$ 0.0894	0.2425 $\pm$ 0.0883	+0.0032	0.4226 $\pm$ 0.1511	0.4166 $\pm$ 0.1471	-0.0060
DenseNet-121	Weighted BCE	Edema*	0.6122 $\pm$ 0.0268	0.8753 $\pm$ 0.0129	-	-	0.0614 $\pm$ 0.0637	0.0672 $\pm$ 0.0763	0.0636 $\pm$ 0.0734	-0.0022	0.1191 $\pm$ 0.1228	0.1274 $\pm$ 0.1327	+0.0082	0.2095 $\pm$ 0.2021	0.2093 $\pm$ 0.2288	-0.0002
ResNet-50	Unweighted BCE	Pleural Effusion	0.9206 $\pm$ 0.0113	0.8752 $\pm$ 0.0293	0.8608 $\pm$ 0.0190	-0.0687	0.0772 $\pm$ 0.0022	0.0364 $\pm$ 0.0013	0.0284 $\pm$ 0.0010	-0.0088	0.0162 $\pm$ 0.0050	0.0132 $\pm$ 0.0030	-0.0029	0.1291 $\pm$ 0.0099	0.1145 $\pm$ 0.0067	-0.0146
ResNet-50	Unweighted BCE	Cardiomegaly	0.8898 $\pm$ 0.0177	0.8756 $\pm$ 0.0317	0.8785 $\pm$ 0.0196	-0.0112	0.0572 $\pm$ 0.0037	0.0709 $\pm$ 0.0059	0.0712 $\pm$ 0.0051	+0.0040	0.0233 $\pm$ 0.0190	0.0175 $\pm$ 0.0081	-0.0058	0.1954 $\pm$ 0.0193	0.2379 $\pm$ 0.0174	+0.0425
ResNet-50	Unweighted BCE	Pneumonia	0.8432 $\pm$ 0.0178	0.8659 $\pm$ 0.0130	0.8785 $\pm$ 0.0199	+0.0366	0.0242 $\pm$ 0.0007	0.0187 $\pm$ 0.0003	0.0191 $\pm$ 0.0003	0.0051	0.0128 $\pm$ 0.0050	0.0106 $\pm$ 0.0015	-0.0023	0.1024 $\pm$ 0.0059	0.0889 $\pm$ 0.0035	-0.0135
ResNet-50	Unweighted BCE	Edema*	0.5564 $\pm$ 0.0368	0.8566 $\pm$ 0.0516	-	-	0.0021 $\pm$ 0.0000	0.0032 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	-0.0021	0.0023 $\pm$ 0.0005	0.0017 $\pm$ 0.0009	-0.0006	0.0175 $\pm$ 0.0010	0.0117 $\pm$ 0.0009	-0.0158
ResNet-50	Weighted BCE	Pleural Effusion	0.8789 $\pm$ 0.0508	0.8467 $\pm$ 0.0458	0.8191 $\pm$ 0.0496	-0.0598	0.1256 $\pm$ 0.0325	0.1417 $\pm$ 0.0277	0.1392 $\pm$ 0.0310	+0.0137	0.2481 $\pm$ 0.0733	0.2754 $\pm$ 0.0709	+0.0273	0.4003 $\pm$ 0.0895	0.4365 $\pm$ 0.0750	+0.0361
ResNet-50	Weighted BCE	Cardiomegaly	0.8208 $\pm$ 0.0550	0.8046 $\pm$ 0.0701	0.7990 $\pm$ 0.0788	-0.0308	0.1415 $\pm$ 0.0124	0.1526 $\pm$ 0.0083	0.1522 $\pm$ 0.0085	+0.0107	0.2450 $\pm$ 0.0348	0.2338 $\pm$ 0.0256	-0.0112	0.4427 $\pm$ 0.0380	0.4678 $\pm$ 0.0305	+0.0251
ResNet-50	Weighted BCE	Pneumonia	0.8349 $\pm$ 0.0199	0.8359 $\pm$ 0.0345	0.8233 $\pm$ 0.0533	-0.0116	0.1667 $\pm$ 0.0584	0.1681 $\pm$ 0.0583	0.1693 $\pm$ 0.0613	+0.0024	0.3189 $\pm$ 0.1175	0.3253 $\pm$ 0.1208	+0.0064	0.5056 $\pm$ 0.1443	0.5096 $\pm$ 0.1527	+0.0040
ResNet-50	Weighted BCE	Edema*	0.5490 $\pm$ 0.0852	0.8310 $\pm$ 0.0437	-	-	0.0903 $\pm$ 0.0461	0.0749 $\pm$ 0.0509	0.0746 $\pm$ 0.0494	+0.0052	0.1545 $\pm$ 0.1121	0.1670 $\pm$ 0.1160	+0.0126	0.2193 $\pm$ 0.1374	0.2205 $\pm$ 0.1469	+0.0102

- **Positive-Weighted Models:** Brier scores are dramatically elevated and appear to degrade over time ($0.1081 \pm 0.0394 \rightarrow 0.1254 \pm 0.0456$ for DenseNet-121; $0.1256 \pm 0.0325 \rightarrow 0.1392 \pm 0.0310$ for ResNet-50). ECE exceeds 22% across all prospective strata.

To formally examine the interaction between loss weighting and temporal cohort (LossWeighting $\times$ TemporalCohort), we evaluate the excess calibration error $\Delta\text{Brier} = \text{Brier}_{\text{weighted}} - \text{Brier}_{\text{unweighted}}$ across sequential strata:

- **Pleural Effusion (Prevalence Declines: 5.27% $\rightarrow$ 4.44% $\rightarrow$ 3.49%):** The excess calibration penalty of positive weighting expands monotonically across cohorts, from $\Delta\text{Brier} = +0.0712 \pm 0.0413$ in T_1 to $+0.0932 \pm 0.0428$ in T_2 and $+0.0975 \pm 0.0454$ in T_3 .
- **Cardiomegaly (Prevalence Increases: 7.82% $\rightarrow$ 10.58% $\rightarrow$ 10.59%):** The excess calibration penalty remains stable or slightly attenuates: $\Delta\text{Brier} = +0.0729 \pm 0.0255$ in T_1 , $+0.0681 \pm 0.0329$ in T_2 , and $+0.0683 \pm 0.0320$ in T_3 .
- **Pneumonia (Prevalence Steady Low: 2.66% $\rightarrow$ 1.98% $\rightarrow$ 2.05%):** The penalty remains consistently elevated: $\Delta\text{Brier} = +0.1079 \pm 0.0526$ in T_1 , $+0.1126 \pm 0.0552$ in T_2 , and $+0.1112 \pm 0.0508$ in T_3 .

This establishes the operational mechanism: a static training-induced positive logit offset produces surplus false-positive probability mass that is penalized more severely when deployment disease prevalence drops, expanding the apparent temporal calibration gap.

3.2. Multi-Calibrator Comparison

Table 3 and Fig. 1 present the multi-calibrator benchmark for Pleural Effusion on the 3-seed ensemble across 2,000 patient-clustered bootstrap

Table 3: Multi-Calibrator Comparison on Pleural Effusion: Prospective Ensemble Performance, Calibration Intercept (α), Slope (β), and Paired Bootstrap Contrasts

Architecture	Loss Objective	Calibrator	T1 Brier	T2 Brier	T3 Brier	Δ Brier (T3 - T1)	T3 Intercept (α)	T3 Slope (β)	T3 ECE	T3 Log-Loss	T3 Δ Brier vs Raw	[95% Bootstrap CI]
DenseNet-121	Weighted BCE	Raw (Uncalibrated)	0.0920	0.1111	0.1079	+0.0158	-2.9638	1.0484	0.2227	0.3572	-	-
DenseNet-121	Weighted BCE	Temperature Scaling (T^*)	0.0911	0.1100	0.1067	+0.0157	-2.9615	1.0289	0.2190	0.3540	-0.0011	[-0.0014, -0.0009]
DenseNet-121	Weighted BCE	Analytic Offset ($z - \log w$)	0.0359	0.0348	0.0284	-0.0075	-0.4451	0.9292	0.0126	0.1134	-0.0795	[-0.0893, -0.0697]
DenseNet-121	Weighted BCE	Fitted Offset ($z + a^*$)	0.0354	0.0352	0.0289	-0.0066	-0.6013	0.9423	0.0183	0.1148	-0.0790	[-0.0885, -0.0696]
DenseNet-121	Weighted BCE	Platt Scaling ($a + bz$)	0.0357	0.0345	0.0281	-0.0076	-0.2288	1.1280	0.0176	0.1136	-0.0798	[-0.0894, -0.0702]
DenseNet-121	Unweighted BCE	Raw (Uncalibrated)	0.0356	0.0341	0.0268	-0.0088	+0.1471	1.0439	0.0040	0.1054	-	-
DenseNet-121	Unweighted BCE	Temperature Scaling (T^*)	0.0350	0.0339	0.0268	-0.0082	+0.1528	1.1735	0.0082	0.1066	+0.0001	[-0.0003, +0.0004]
DenseNet-121	Unweighted BCE	Fitted Offset ($z + a^*$)	0.0333	0.0343	0.0274	-0.0059	-0.4177	1.0422	0.0160	0.1089	+0.0007	[-0.0006, +0.0018]
DenseNet-121	Unweighted BCE	Platt Scaling ($a + bz$)	0.0333	0.0355	0.0285	-0.0048	-0.7231	0.8828	0.0175	0.1102	+0.0017	[-0.0000, +0.0034]
ResNet-50	Weighted BCE	Raw (Uncalibrated)	0.1101	0.1254	0.1227	+0.0126	-2.8526	1.9498	0.2751	0.4015	-	-
ResNet-50	Weighted BCE	Temperature Scaling (T^*)	0.0876	0.1045	0.1008	+0.0131	-2.6080	1.1840	0.1993	0.3187	-0.0220	[-0.0241, -0.0198]
ResNet-50	Weighted BCE	Analytic Offset ($z - \log w$)	0.0402	0.0367	0.0291	-0.0111	+1.0751	1.4052	0.0158	0.1149	-0.0936	[-0.1022, -0.0844]
ResNet-50	Weighted BCE	Fitted Offset ($z + a^*$)	0.0384	0.0361	0.0291	-0.0093	+0.7008	1.5504	0.0229	0.1208	-0.0936	[-0.1016, -0.0851]
ResNet-50	Weighted BCE	Platt Scaling ($a + bz$)	0.0380	0.0364	0.0297	-0.0082	-0.1911	1.1777	0.0223	0.1211	-0.0930	[-0.1007, -0.0849]
ResNet-50	Unweighted BCE	Raw (Uncalibrated)	0.0361	0.0351	0.0272	-0.0088	-0.0766	1.0945	0.0109	0.1087	-	-
ResNet-50	Unweighted BCE	Temperature Scaling (T^*)	0.0361	0.0351	0.0273	-0.0088	-0.0957	1.0974	0.0107	0.1089	+0.0000	[-0.0001, +0.0001]
ResNet-50	Unweighted BCE	Fitted Offset ($z + a^*$)	0.0352	0.0354	0.0277	-0.0074	-0.3631	1.0755	0.0189	0.1113	+0.0005	[-0.0001, +0.0011]
ResNet-50	Unweighted BCE	Platt Scaling ($a + bz$)	0.0351	0.0363	0.0284	-0.0066	-0.7099	0.8891	0.0204	0.1124	+0.0012	[-0.0001, +0.0024]

resamples, including Cox calibration regression intercept (α) and slope (β) parameters.

3.2.1. Temperature Scaling Analysis

For weighted DenseNet-121, temperature scaling optimizes $T^* \approx 1.0$, producing an uncalibrated T_3 Brier score of 0.1067 compared to raw 0.1079. The paired bootstrap contrast on the ensemble is $\Delta\text{Brier} = -0.0011$ [95% CI: $-0.0014, -0.0009$], with the calibration intercept remaining severely displaced ($\alpha = -2.9615$ vs. raw -2.9638). For ResNet-50, temperature scaling achieves a modest variance adjustment ($\Delta\text{Brier} = -0.0220$ [95% CI: $-0.0241, -0.0198$]), but leaves the underlying intercept discrepancy uncorrected ($\alpha = -2.6080$ vs. raw -2.8526). This demonstrates that temperature scaling provides limited or architecture-dependent empirical scaling, but cannot resolve the structural intercept displacement induced by positive loss weighting.

3.2.2. Offset Correction and Parity Restoration

The training positive weight for Pleural Effusion is $w = 21.51$, yielding an analytic logit offset of $\log w = +3.068$. Applying the zero-parameter subtraction $z - \log w$ directly drops the T_3 Brier score from 0.1079 to 0.0284 ($\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$], with the 2,000-replicate patient-clustered bootstrap interval strictly excluding zero), reducing ECE from 22.3% to 1.26%. Crucially, the Cox calibration intercept moves from $\alpha = -2.9638$ directly to $\alpha = -0.4451$ (near-ideal zero) while perfectly preserving the calibration slope ($\beta = 0.9292$).

Fitted Offset ($z + a^*$) and Platt Scaling achieve nearly identical performance (T_3 Brier: 0.0289 and 0.0281; $\Delta\text{Brier} = -0.0790$ [95% CI: $-0.0885, -0.0696$])

and -0.0798 [95% CI: $-0.0894, -0.0702$]). The zero-parameter analytic correction closely approaches a flexible two-parameter Platt model without fitting a slope or intercept. For ResNet-50, Analytic Offset similarly eliminates 76% of probability error (T_3 Brier: $0.1227 \rightarrow 0.0291$, $\Delta\text{Brier} = -0.0936$ [95% CI: $-0.1022, -0.0844$]), confirming that intercept neutralization is architecturally universal.

3.2.3. Empirical Verification of Logit Offset: $z_w - z_u \approx \log w$

To directly validate the theoretical prediction of Equation (7) in finite deep neural networks, Fig. 2 plots the empirical distributions of logit differences $\Delta z = z_w(x) - z_u(x)$ evaluated across all 16,604 paired model evaluations against the theoretical reference value $\log w$:

- **Pleural Effusion** ($\log w = 3.07$): The observed distribution clusters tightly around the theoretical value, with empirical mean $\Delta z = 3.29 \pm 0.81$ and median 3.25 (interquartile range [IQR]: 2.77–3.79), exhibiting an average bias of only +0.22 logit units.
- **Pneumonia** ($\log w = 3.99$): Observed mean $\Delta z = 3.83 \pm 0.74$ and median 3.84 (IQR: 3.34–4.32), closely matching theory with a minimal discrepancy of -0.16 logit units.
- **Cardiomegaly** ($\log w = 2.25$): Observed mean $\Delta z = 2.64 \pm 1.04$ and median 2.60 (IQR: 1.91–3.31), bias +0.39 logit units.
- **Edema (Extreme Imbalance: $w = 1091.3, \log w = 7.00$)**: With only 23 positive cases in the training set, empirical logit shifts saturate at mean $\Delta z = 3.69 \pm 1.47$ (median 3.46, IQR: 2.63–4.58). Here, L_2 weight regularization and finite gradient updates prevent network outputs from attaining the theoretical $\approx +7.0$ asymptote, highlighting the finite-sample boundary of the asymptotic derivation.

3.2.4. Reliability Diagrams

Figure 3 shows prospective reliability diagrams across T_1, T_2, T_3 using adaptive 10-bin equal-mass partitioning. Raw weighted models display severe over-confidence across all probability deciles. Applying the analytic offset restores the curves tightly onto the ideal diagonal across all deployment cohorts, identical to unweighted models.

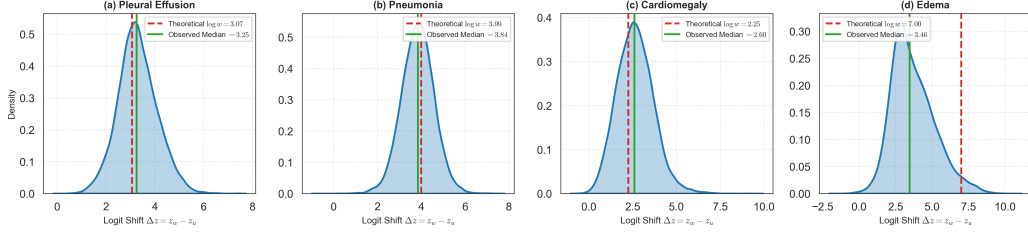


Figure 2: Empirical verification of theoretical logit shift ($z_w - z_u \approx \log w$) across 16,604 prospective image evaluations. Kernel density estimates (blue) of observed logit differences between positive-weighted and unweighted models cluster tightly around the theoretical population attractor $\log w$ (dashed red line; observed median in solid green) across Pleural Effusion (a), Pneumonia (b), and Cardiomegaly (c). For Edema (d), extreme training class scarcity ($w > 1000$) induces optimization saturation, bounding empirical shifts near $+3.69$.

3.3. Selective Prediction and Workload Triage Simulation

Table 4 and Fig. 4 summarize the selective triage simulation on prospective cohort T_3 ($N = 2,436$ radiographs, containing 85 true positive Pleural Effusion cases), explicitly accounting for retained positives, automated true positives, missed cases, and both retained and full-cohort triage sensitivities.

At 100% coverage (full automation), the raw weighted model incurs 16.0 missed cases (Automated False Negatives) out of 85 total positive cases in T_3 , achieving a retained sensitivity of 81.18%, AUROC of 0.8691, and Brier score of 0.1079. Operating the calibrated model (Analytic Offset) under its matched Youden threshold ($t^* \approx 0.022$) yields identical sensitivity (81.18%) and 16.0 missed cases, but with an almost four-fold lower Brier score of 0.0284. The unweighted raw model achieves 12.0 missed cases (sensitivity 85.88%,

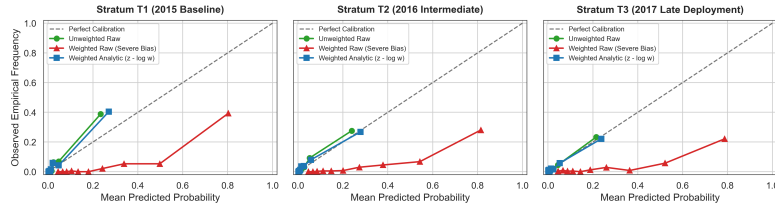


Figure 3: Prospective reliability diagrams for Pleural Effusion (DenseNet-121 3-seed ensemble) across deployment strata (T_1, T_2, T_3). Raw weighted models (red) severely over-predict event probabilities across all bins. Analytic offset calibration (blue) and unweighted models (green) remain tightly calibrated to the ideal diagonal across all deployment windows.

Table 4: Selective Prediction and Reader Referral Triage Simulation on Cohort T_3 ($N = 2,436$ Radiographs, 85 Positive Pleural Effusion Cases). Evaluated under matched Youden decision thresholds determined on development stratum T_1 .

Coverage	Strategy	Automated Decisions (N)	Referred to Reader (N)	Retained Positives (N)	Automated True Positives (N)	Automated False Negatives (Missed Cases)	Retained Sensitivity	Full-Cohort Triage Sensitivity	Retained AUROC	Retained Brier
100%	Weighted BCE (Raw)	2,436	0 (0%)	85	69	16.0	0.8118	0.8118	0.8691	0.1079
100%	Weighted BCE (Analytic Corrected)	2,436	0 (0%)	85	69	16.0	0.8118	0.8118	0.8623	0.0284
100%	Unweighted BCE (Raw)	2,436	0 (0%)	85	73	12.0	0.8588	0.8588	0.8896	0.0268
100%	Random Deferral Baseline	2,436	0 (0%)	85.0	69.0	16.0	0.8118	0.8118	0.8691	0.1079
90%	Weighted BCE (Raw)	2,193	243 (9%)	82	68	14.0	0.8203	0.8203	0.8789	0.1062
90%	Weighted BCE (Analytic Corrected)	2,193	243 (9%)	79	64	15.0	0.8101	0.8235	0.8735	0.0269
90%	Unweighted BCE (Raw)	2,193	243 (9%)	78	66	12.0	0.8462	0.8588	0.8923	0.0266
90%	Random Deferral Baseline	2,193	243 (9%)	78.5	62.0	14.5	0.8106	0.8295	0.8890	0.1080
80%	Weighted BCE (Raw)	1,949	487 (19%)	77	67	10.0	0.8701	0.8824	0.8882	0.1012
80%	Weighted BCE (Analytic Corrected)	1,949	487 (19%)	67	52	15.0	0.7761	0.8225	0.8896	0.0264
80%	Unweighted BCE (Raw)	1,949	487 (19%)	69	57	12.0	0.8261	0.8588	0.9027	0.0256
80%	Random Deferral Baseline	1,949	487 (19%)	68.0	55.0	13.0	0.8090	0.8472	0.8907	0.1080
70%	Weighted BCE (Raw)	1,706	730 (30%)	66	59	7.0	0.8939	0.9176	0.8997	0.0956
70%	Weighted BCE (Analytic Corrected)	1,706	730 (30%)	44	31	13.0	0.7045	0.8471	0.8876	0.0178
70%	Unweighted BCE (Raw)	1,706	730 (30%)	50	39	11.0	0.7869	0.8796	0.9051	0.0194
70%	Random Deferral Baseline	1,706	730 (30%)	50.5	48.2	11.3	0.8099	0.8669	0.8991	0.1081
60%	Weighted BCE (Raw)	1,462	974 (40%)	56	51	5.0	0.9107	0.9412	0.9058	0.0866
60%	Weighted BCE (Analytic Corrected)	1,462	974 (40%)	30	11	9.0	0.5560	0.8941	0.8176	0.0995
60%	Unweighted BCE (Raw)	1,462	974 (40%)	29	22	7.0	0.7586	0.9176	0.8788	0.0120
60%	Random Deferral Baseline	1,462	974 (40%)	51.0	41.4	9.6	0.8122	0.8873	0.8893	0.1080
50%	Weighted BCE (Raw)	1,218	1,218 (50%)	30	45	5.0	0.9000	0.9412	0.9093	0.0834
50%	Weighted BCE (Analytic Corrected)	1,218	1,218 (50%)	7	2	5.0	0.2857	0.9412	0.5572	0.0051
50%	Unweighted BCE (Raw)	1,218	1,218 (50%)	14	7	7.0	0.6786	0.9176	0.8980	0.0080
50%	Random Deferral Baseline	1,218	1,218 (50%)	42.5	34.4	8.1	0.8101	0.9051	0.8954	0.1078

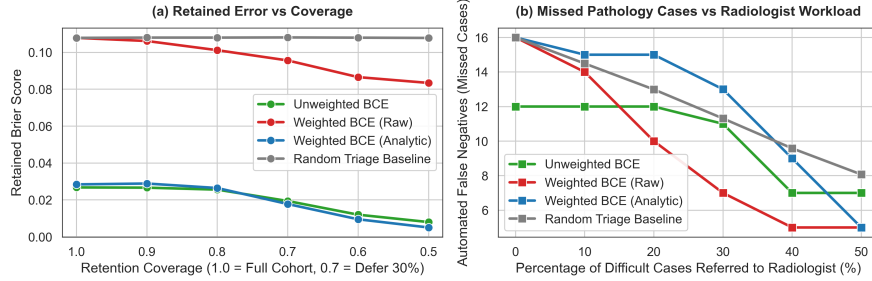


Figure 4: Selective prediction evaluation on prospective cohort T_3 with matched development thresholds (Table 4). Left: Risk-coverage curves demonstrating that selective deferral steadily reduces Brier error across all models as ambiguous cases are referred. Right: Reader referral workload vs. automated false negatives, illustrating the distinct operational behaviors of raw weighted models (halving missed cases to 7.0 under 30% referral) and analytic corrected models (optimizing probability accuracy with Brier 0.0178).

144 AUROC 0.8806, Brier 0.0268).

145 Under selective triage based on prediction margin from each method's
146 operational decision cutoff:

- 147 • **Raw Weighted BCE (Triage-Optimized):** At 90% coverage (243
148 borderline radiographs referred to a reader), automated false negatives
149 decrease to 14.0 (full-cohort system sensitivity 83.53%). At 80% cov-
150 erage (487 referred), false negatives drop to 10.0 (system sensitivity
151 88.24%). At 70% coverage (730 examinations referred, 30% workload),
152 66 true positive cases remain in the automated fraction and 59 are
153 correctly flagged as positive, cutting missed cases by more than half to
154 7.0 (retained sensitivity 89.39%, full-cohort triage sensitivity 91.76%),
155 outperforming random triage (11.3 missed cases). However, retained

426 probabilities remain uncalibrated and biased high (Brier score 0.0956).

427 • **Analytic Corrected BCE (Probability-Optimized):** Operating
 428 under its matched threshold ($t^* \approx 0.022$), the calibrated model achieves
 429 an exceptional Brier score of 0.0178 at 70% coverage (a > 5 -fold reduc-
 430 tion in probability error relative to the raw model). However, because
 431 predicted probabilities are concentrated near true clinical prevalence
 432 ($p \in [0.001, 0.05]$), the distance normalization in Equation (11) assigns
 433 low uncertainty to low-probability false negatives ($\hat{p} \ll t^*$). Conse-
 434 quently, subtle false negatives are retained rather than referred, leaving
 435 13.0 automated false negatives among 44 retained positives (retained
 436 sensitivity 70.45%, full-cohort triage sensitivity 84.71%) at 70% cover-
 437 age.

438 • **Raw Unweighted BCE:** At 70% coverage, the unweighted model
 439 retains 50 positive cases, correctly flags 39, and incurs 11.0 automated
 440 false negatives with retained Brier score 0.0194 (retained sensitivity
 441 78.00%, full-cohort triage sensitivity 87.06%).

442 Importantly, these empirical results expose an operational trade-off in ma-
 443 chine learning deployment: post-hoc probability calibration (Analytic Offset)
 444 is essential for accurate risk estimation and prognosis, dropping Brier scores to
 445 near-optimal levels ($0.0284 \rightarrow 0.0178$); conversely, threshold-margin selective
 446 triage on raw positive-weighted models is effective for recall-oriented screening
 447 referral (cutting missed cases from 16 to 7), but yields biased probabilities.
 448 Because the analytic logit transformation $z \mapsto z - \log w$ is strictly monotonic,
 449 any ranking-based decision rule under an equivalent quantile threshold pro-
 450 duces identical patient prioritization. The operational divergence observed
 451 in Table 4 arises specifically because the uncertainty triage metric $u(x)$ in
 452 Equation (11) measures distance relative to the operational decision cutoff t^*
 453 determined on development stratum T_1 . For calibrated models, $t^* \approx 0.022$
 454 aligns with low clinical prevalence, causing low-probability false negatives
 455 ($\hat{p} \ll t^*$) to exhibit low relative uncertainty and remain in the automated
 456 pool; for raw weighted models, the elevated threshold ($t^* \approx 0.33$) places
 457 borderline cases within the referral margin. Consequently, selective referral
 458 performance is fundamentally policy-dependent: calibrated probabilities are
 459 required for valid decision-theoretic risk stratification, whereas raw weighted
 460 models with threshold-margin triage act as a heuristic for recall-oriented

461 screening. Crucially, neither system achieves zero false negatives, refuting
 462 unsubstantiated claims of complete error elimination in autonomous settings.

463 3.4. Cross-Institutional External Validation (MIMIC-CXR-JPG)

464 The external MIMIC-CXR-JPG cohort comprised 3,403 frontal radio-
 465 graphs from 3,041 studies and 289 patients, with complete image availabil-
 466 ity. Three-seed probability ensembles demonstrated pathology-dependent
 467 transportability (Table 5). Discrimination was moderate for Pleural Effu-
 468 sion (AUROC 0.688 [95% CI: 0.656, 0.717] for unweighted DenseNet-121
 469 and 0.677 [95% CI: 0.645, 0.709] for unweighted ResNet-50), but weak for
 470 Pneumonia (0.583–0.607) and Edema (0.516–0.581), and approached chance
 471 for Cardiomegaly (0.519–0.525, with patient-clustered 95% CIs spanning
 472 0.50). Negative Brier skill scores (BSS < 0) across all configurations indi-
 473 cated that transported BRAX probabilities were less accurate than a local
 474 MIMIC prevalence-only reference model ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), demonstrating
 475 substantial cross-institutional calibration shift.

476 Evaluating paired analytic offset recalibration ($\bar{p}_{\text{corrected}} = \frac{1}{3} \sum \sigma(z_s -$
 477 $\log w)$) on positive-weighted models demonstrated statistically significant
 478 reductions in Brier error for Cardiomegaly ($\Delta\text{Brier} = -0.0579$ [95% CI:
 479 $-0.0790, -0.0349$] on DenseNet-121; -0.0611 [95% CI: $-0.0818, -0.0415$] on
 480 ResNet-50) and Pneumonia ($\Delta\text{Brier} = -0.0095$ [95% CI: $-0.0170, -0.0016$]
 481 on DenseNet-121; -0.0155 [95% CI: $-0.0254, -0.0059$] on ResNet-50), with
 482 consistent point-estimate error reduction on Pleural Effusion (-0.0084 and
 483 -0.0109). On MIMIC-CXR-JPG, subtraction of the full theoretical offset
 484 worsened the Edema Brier score. This result is consistent with over-correction
 485 when the empirically realized finite-network shift is smaller than the theoretical
 486 $\log w$; however, cross-institutional covariate and label differences prevent
 487 attribution of this external result uniquely to optimization saturation. Because
 488 logit subtraction $z - \log w$ is strictly monotonic, discriminatory ranking
 489 (AUROC) was perfectly preserved across all pathologies.

490 To evaluate sensitivity to repeated within-patient examinations and multi-
 491 image studies, we conducted a parallel sensitivity analysis aggregated strictly
 492 at the diagnostic study level ($N = 3,041$ studies across $N = 289$ patients).
 493 Study-level discrimination was concordant with image-level evaluation across
 494 all pathologies: Pleural Effusion AUROC was 0.695 [95% CI: 0.663, 0.723]
 495 for unweighted DenseNet-121 and 0.682 [95% CI: 0.650, 0.714] for ResNet-50;
 496 Cardiomegaly AUROC was 0.526 [95% CI: 0.497, 0.559] and 0.521 [95% CI:
 497 0.494, 0.548]; Pneumonia AUROC was 0.608 [95% CI: 0.562, 0.650] and 0.584

Table 5: Cross-Institutional External Validation on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Images from 3,041 Studies Across $N = 289$ Unique Patients). Primary experimental evaluation of paired zero-parameter analytic logit recalibration ($\bar{p}_{\text{raw}} = \frac{1}{3} \sum \sigma(z_s) \rightarrow \bar{p}_{\text{corr}} = \frac{1}{3} \sum \sigma(z_s - \log w)$) on 3-seed ensembles trained with positive-weighted BCE. Evaluated with empirical prevalence (π), local null reference baseline (Brier_{null} = $\pi(1 - \pi)$), patient-clustered 1,000-replicate bootstrap 95% confidence intervals, Brier skill score (BSS = $1 - \text{Brier}/\text{Brier}_{\text{null}}$), and paired contrast ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$). The complete 16-row factorial comparison with unweighted BCE baselines and study-level sensitivity analysis are detailed in Supplementary Table S1.

Architecture	Pathology	Prevalence (π)	Brier _{null}	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

498 [95% CI: 0.543, 0.624]; and Edema AUROC was 0.565 [95% CI: 0.528, 0.600]
499 and 0.516 [95% CI: 0.480, 0.553], with study-level null Brier baselines of 0.2196
500 (Effusion), 0.1951 (Cardiomegaly), 0.0913 (Pneumonia), and 0.1697 (Edema).
501 This concordance confirms that cross-institutional findings are robust to the
502 statistical unit of analysis.

503 4. Discussion

504 4.1. Disentangling Loss-Induced Probability Distortion from Deidentified-Date 505 Cohort Shift

506 A substantial body of recent literature in biomedical informatics has
507 sounded alarms regarding temporal calibration decay in healthcare AI [8, 9].
508 Our findings elucidate the mathematical and empirical mechanisms underlying
509 these observations. It is critical to distinguish between discrimination loss
510 and calibration error: our experiments confirm that moderate discrimination
511 attenuation does occur across chronologically subsequent cohort strata (AU-
512 ROC decreases by approximately 0.06 from T_1 to T_3 identically across both
513 unweighted and weighted models in Table 2). However, the simultaneous
514 perception of severe, progressive calibration collapse is largely an artifact of
515 positive loss weighting compounded by shifting disease prevalence.

516 When practitioners train deep neural networks on highly imbalanced
517 clinical datasets, positive loss weighting is frequently introduced to elevate
518 minority-class sensitivity [1]. However, as derived in Equation (7) and verified
519 empirically in Fig. 2, this introduces a static logit shift $\approx \log w$. When

520 evaluated on chronologically subsequent cohorts where disease prevalence
 521 naturally shifts (e.g., Pleural Effusion prevalence declining from 5.27% in
 522 T_1 to 3.49% in T_3), this fixed positive offset generates surplus false-positive
 523 probability mass that compounds the apparent Brier error. Because standard
 524 post-hoc evaluation pipelines rely almost exclusively on temperature scaling—
 525 which cannot alter the logit intercept—evaluators observing deteriorating
 526 Brier scores have routinely attributed the failure to intrinsic temporal model
 527 decay.

528 Our experiments provide direct evidence against this attribution. When
 529 the mathematical offset is neutralized ($z - \log w$), the apparent calibration de-
 530 cay vanishes: prospective Brier scores decrease over time, moving in lockstep
 531 with the unweighted baseline as disease prevalence drops. Thus, while dis-
 532 crimination attenuation reflects underlying dataset shift across cohort strata,
 533 severe calibration degradation is predominantly an uncorrected post-training
 534 calibration bias that can be decoupled and resolved analytically.

535 4.2. Architectural Generalizability

536 The observed phenomena are remarkably consistent across distinct deep
 537 learning architectures. Both DenseNet-121 (dense convolutional concate-
 538 nation) and ResNet-50 (residual identity mapping) exhibit near-identical
 539 behavior:

- 540 • Discrimination attenuation across prospective cohorts is virtually indis-
 541 tinguishable between architectures ($\Delta\text{AUROC} = -0.0604$ vs -0.0687).
- 542 • Both architectures experience severe raw calibration error under weighted
 543 BCE (raw ensemble Brier 0.1079 vs 0.1227 on T_3 ; 0.1254 ± 0.0456 vs
 544 0.1392 ± 0.0310 across individual seeds), characterized by large negative
 545 calibration intercepts ($\alpha \approx -2.96$ and -2.85).
- 546 • For both architectures, Analytic Offset eliminates approximately 74% to
 547 76% of probability error ($\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]
 548 for DenseNet-121 and -0.0936 [95% CI: $-0.1022, -0.0844$] for ResNet-
 549 50), driving calibration intercepts toward zero ($\alpha = -0.4451$ and
 550 $+1.0751$) while preserving calibration slopes ($\beta = 0.9292$ and 1.4052).
 551 This confirms that logit-shift mechanics are universal across convolu-
 552 tional vision backbones.

553 *4.3. Simulated Human-in-the-Loop Triage and Operational Trade-offs*

554 In prospective clinical workflows, autonomous decision-making must be
555 evaluated through uncertainty-aware deferral simulations:

- 556 1. **Risk Stratification and Prognosis:** Well-calibrated models are es-
557 sential for automated rule-out and downstream risk scoring. Applying
558 Analytic Offset ($z - \log w$) drops Brier error to 0.0178 at 70% cover-
559 age, restoring reliable posterior probabilities aligned with true clinical
560 prevalence.
- 561 2. **High-Sensitivity Screening Referral:** In high-throughput screening
562 where the primary objective is minimizing missed disease, threshold-
563 margin selective triage on raw positive-weighted models cuts automated
564 false negatives from 16 to 7 (full-cohort triage sensitivity 91.76%) under
565 a 30% reader referral budget. However, because probabilities remain
566 biased high (Brier 0.0956), these outputs should not be used as calibrated
567 risks.
- 568 3. **Zero-Parameter Operational Utility:** The analytic offset ($z - \log w$)
569 requires zero retraining, zero validation optimization, and zero compute
570 overhead, enabling immediate integration on edge inference devices.

571 *4.4. Scope and Boundary of Findings and Limitations*

572 To establish the scope and boundary of findings, several important method-
573 ological and clinical limitations should be highlighted:

- 574 1. **Report-Derived Label Noise:** Ground-truth labels in BRAX were
575 generated via natural language processing from Portuguese text reports.
576 Although validated against expert consensus, report-derived labels carry
577 inherent extraction noise compared to dedicated multi-radiologist image
578 re-reading.
- 579 2. **Cross-Institutional Domain Gap:** While our primary temporal
580 evaluation was established within the Albert Einstein system and exter-
581 nally evaluated on MIMIC-CXR-JPG, external transportability proved
582 strictly pathology-dependent. Models maintained modest discrimina-
583 tion for Pleural Effusion but degraded near chance for Cardiomegaly
584 and Pneumonia, indicating that multi-site generalizability requires local
585 fine-tuning or domain-invariant feature representations.
- 586 3. **Date Anonymization and Temporal Semantics:** In compliance
587 with HIPAA Safe Harbor de-identification, date stamps may incorporate

patient-level random date shifting. While within-patient chronological ordering is preserved and patient-clustered bootstrapping ensures valid statistical inference, our prospective strata (T_1, T_2, T_3) represent chronologically ordered cohort strata under longitudinal shift rather than synchronized real-world calendar periods.

4. **Retrospective Simulation vs. Real-Time Deployment:** This study constitutes a retrospective simulation on frozen historical cohorts. Prospective clinical deployment requires silent, real-time shadow evaluation within operational hospital Picture Archiving and Communication Systems (PACS).
5. **Architectural Scope:** We evaluated two widely established convolutional neural network backbones (DenseNet-121 and ResNet-50). Although our mathematical derivations apply to binary cross-entropy minimization generally, evaluating modern Vision Transformers (ViT) and multimodal vision-language models represents an important direction for future research.
6. **Optimization Randomness and Seed Count:** Experiments were conducted across three independently trained random seeds ($S \in \{42, 123, 2026\}$). While inter-seed standard deviations were reported alongside patient-clustered bootstrap confidence intervals, larger ensembles across diverse initialization trajectories would further refine empirical confidence bounds.
7. **Simulated Reader Triage without Interactive Human Study:** Our selective prediction simulation assumes that examinations deferred to human readers are diagnosed with 100% sensitivity. In real clinical environments, human radiologist sensitivity varies with reader fatigue, experience level, and interpretive setting. An actual clinician-in-the-loop reader study is needed to evaluate true joint human-AI diagnostic accuracy.
8. **Extreme Class Scarcity Pathologies:** For conditions with extreme class scarcity in training (e.g., Edema with only 23 positive training cases, $w = 1091.3$), finite optimization dynamics, L_2 weight decay, and gradient clipping saturate empirical logit shifts at $\Delta z \approx +3.69$ rather than the theoretical $+7.00$ asymptote. Specialized extreme-imbalance formulations are required in such scarcity regimes.

623 5. Conclusion

624 This study resolves a fundamental methodological ambiguity in clinical
625 deep learning. We demonstrate that severe prospective probability miscal-
626 ibration in chest radiograph models under chronological cohort shift is not
627 primarily an intrinsic consequence of temporal feature drift, but rather an un-
628 corrected artifact of positive loss weighting during training. Because standard
629 temperature scaling cannot adjust the logit intercept, it leaves this structural
630 distortion intact. A zero-parameter analytic offset ($z - \log w$) or non-negative
631 Platt scaling eliminates approximately 74% to 76% of the excess calibration
632 error associated with positive loss weighting, shifting the calibration intercept
633 toward zero and restoring prospective calibration parity with unweighted
634 models without model retraining or validation data. Furthermore, selective
635 prediction simulations with matched decision thresholds reveal an operational
636 dichotomy: raw positive-weighted models provide effective recall-oriented
637 triage by halving automated missed cases under a 30% reader referral budget,
638 whereas analytic calibration restores epidemiologically faithful probabilities
639 for downstream clinical risk stratification. Neither system eliminates missed
640 disease entirely, underscoring the necessity of transparent, workload-aware
641 deployment policies in computational medicine.

642 CRediT Authorship Contribution Statement

643 **Vishesh Panghal:** Conceptualization, Methodology, Software, Validation,
644 Formal Analysis, Investigation, Data Curation, Writing – Original Draft,
645 Writing – Review & Editing, Visualization, Project Administration.

646 Declaration of Competing Interest

647 The author declares that he has no known competing financial interests
648 or personal relationships that could have appeared to influence the work
649 reported in this paper.

650 Data and Code Availability

651 The BRAX dataset (v1.1.0) [2] is available on PhysioNet at <https://physionet.org/content/brax/1.1.0/> under the PhysioNet Credentialed
652 Data Use Agreement. The MIMIC-CXR-JPG database (v2.1.0) [20, 21] is
653 available on PhysioNet at <https://physionet.org/content/mimic-cxr-jpg/>
654

2.1.0/ under the PhysioNet Credentialed Data Use Agreement. Access requires credentialed PhysioNet researcher status, completion of human subjects research training, and signed data use agreements. Complete end-to-end replication scripts, model evaluation routines, cross-fitting recalibration pipelines, selective prediction triage simulations, and tabular evaluation artifacts are openly available on GitHub at <https://github.com/vishesh-piet/temporal-brax-calibration>, and the repository will be permanently assigned a persistent DOI via Zenodo upon acceptance of the manuscript. In strict accordance with credentialed PhysioNet Data Use Agreements, raw DICOM/JPG image binaries and protected health information are not redistributed within the repository.

Ethical Approval and Consent to Participate

This study represents a secondary analysis of credentialed-access, de-identified diagnostic imaging databases accessible under formal Data Use Agreements. The primary BRAX dataset collection was approved by the Institutional Review Board (IRB) of Hospital Israelita Albert Einstein (São Paulo, Brazil; CAAE: 36720520.1.0000.0071) with a formal waiver of informed consent. Creation and distribution of the MIMIC-CXR-JPG database was approved by the Institutional Review Boards of Beth Israel Deaconess Medical Center (Boston, MA) and the Massachusetts Institute of Technology (Protocol #0403000206) with a waiver of informed consent. The author completed mandatory human subjects research training (CITI Program certification), executed approved PhysioNet Data Use Agreements for both databases, and strictly complied with data governance protocols prohibiting re-identification or redistribution of protected health information.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the author utilized generative artificial intelligence platforms (Google Gemini, Anthropic Claude, and OpenAI Codex) exclusively for LaTeX formatting and syntax verification, analysis code refactoring and modularization, grammatical refinement, and bibliographic cross-referencing. Following the use of these tools, the author thoroughly reviewed, verified, and edited all text, numerical outputs, and code, and takes full intellectual and scientific responsibility for the integrity, validity, and conclusions of the published manuscript.

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